

Paper 1 - Set 04 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	A	Quadratic functions
2	B	Differentiation
3	D	Number, units and estimation
4	C	Integration
5	F	Sequences and series
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Worked solutions

Question 1 QUA-18 · Quadratic functions · spec MM1.3

Find the range of values of x for which $x^2 - 7x + 10 < 0$ and $x^2 - 4 > 0$.

Answer: A

$(x - 2)(x - 5) < 0$ gives $2 < x < 5$; $x^2 > 4$ gives $x < -2$ or $x > 2$. The intersection is $2 < x < 5$.

Full method - how and why

Two simultaneous quadratic inequalities. Solve each separately, then intersect on a number line.

- 1 First: $x^2 - 7x + 10 < 0$ factorises as $(x - 2)(x - 5) < 0$, giving $2 < x < 5$.

why: An upward parabola is **below** the axis between its roots.

- 2 Second: $x^2 - 4 > 0$ gives $x < -2$ or $x > 2$.

why: Above the axis means outside the roots - two intervals.

- 3 Intersect: $(2, 5)$ overlaps $(2, \infty)$ entirely and misses $(-\infty, -2)$ completely, so the answer is $2 < x < 5$.

why: 'And' means intersection: a value must satisfy both, so draw both solution sets on one line and take the common part.

Where it goes wrong

- Taking the union and including $x < -2$ - the word is 'and', not 'or'.
- Getting the first inequality backwards ($x < 2$ or $x > 5$) by forgetting that < 0 means between the roots.

Takeaway < 0 between the roots, > 0 outside them (for an upward parabola); then combine solution sets on a number line.

Question 2 DIF-04 · Differentiation · spec MM6.3

Find the equation of the normal to the curve $y = \sqrt{x}$ at the point $(4, 2)$.

Answer: B

$y' = \frac{1}{2\sqrt{x}} = \frac{1}{4}$ at $x = 4$, so the normal has gradient -4 : $y - 2 = -4(x - 4)$, i.e. $y = -4x + 18$.

Full method - how and why

Normal line: the negative reciprocal of the tangent's gradient, through the same point.

- 1 Write $y = x^{1/2}$ and differentiate: $y' = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$.

why: Convert the root to a fractional power before differentiating.

- 2 At $x = 4$: $y' = \frac{1}{2 \times 2} = \frac{1}{4}$.

why: $\sqrt{4} = 2$.

- 3 Normal gradient $= -\frac{1}{1/4} = -4$.

why: Normal is perpendicular to the tangent: negative reciprocal.

- 4 Line through $(4, 2)$: $y - 2 = -4(x - 4) \Rightarrow y = -4x + 18$.

why: $-4 \times -4 = +16$, plus the 2.

Where it goes wrong

- Using the tangent gradient $\frac{1}{4}$ and choosing $y = \frac{1}{4}x + 1$.
- Arithmetic on the constant: $y = -4x + 16$ is offered.

Takeaway Normal gradient $= -1/f'(x_1)$; then substitute the point into the point-gradient form.

Question 3 NUM-01 · Number, units and estimation · spec M2.3

Find the highest common factor of $2^3 \times 3^2 \times 5$ and $2^2 \times 3^3 \times 7$.

Answer: D

Take the lower power of each common prime: $2^2 \times 3^2 = 36$.

Full method - how and why

HCF from prime factorisations: take each **common** prime to its **lower** power.

- 1 List the primes in each: $2^3 \times 3^2 \times 5$ and $2^2 \times 3^3 \times 7$.

why: Both are already factorised, which is the form the rule needs.

- 2 Take the common primes only: 2 and 3 appear in both; 5 and 7 do not.

why: A prime missing from one number cannot divide both.

- 3 Take the lower power of each: $2^{\min(3,2)} = 2^2$ and $3^{\min(2,3)} = 3^2$.

why: The HCF must divide both, so it cannot use more copies of a prime than the smaller number has.

- 4 Multiply: $4 \times 9 = 36$.

why: Check: 36 divides 360 and 756 ✓.

Where it goes wrong

- Taking the higher powers (that gives the LCM contribution, not the HCF).
- Including 5 or 7, which gives a number dividing only one of them.

Takeaway HCF: common primes, lowest powers. LCM: all primes, highest powers.

Question 4 INT-17 · Integration · spec MM7.6

The gradient of a curve at (x, y) is $3\sqrt{x}$ and the curve passes through $(4, 20)$. Find the y-intercept of the curve.

Answer: C

$y = 2x^{3/2} + c$; $20 = 16 + c \Rightarrow c = 4$. The y-intercept is 4.

Full method - how and why

Integrate the gradient function, use the given point for c , then read the intercept.

- 1 Integrate: $\int 3x^{1/2} dx = 3 \times \frac{x^{3/2}}{3/2} = 2x^{3/2}$, so $y = 2x^{3/2} + c$.

why: $3 \div \frac{3}{2} = 2$.

- 2 Substitute $(4, 20)$: $4^{3/2} = 8$, so $2(8) + c = 20$, giving $c = 4$.

why: $4^{3/2} = (\sqrt{4})^3 = 8$.

- 3 The y-intercept is the value at $x = 0$: $y = 0 + 4 = 4$.

why: For this curve the intercept is just c .

Where it goes wrong

- Using $4^{3/2} = 6$ and getting $c = 8$ - an offered option.
- Reporting 20 (the given y-value) as the intercept.

Takeaway The constant of integration **is** the y-intercept when the rest of the expression vanishes at $x = 0$.

Question 5 SEQ-15 · Sequences and series · spec MM2.3

Evaluate $\sum_{k=1}^{10} 3 \times 2^k$.

Answer: F

This is $6 + 12 + \dots + 6 \cdot 2^9$: first term 6, ratio 2, 10 terms: $6(2^{10} - 1) = 6 \times 1023 = 6138$.

Full method - how and why

Sigma notation hiding a geometric series. Write out the first term and the ratio, and count the terms carefully.

- 1 Write out the start: $k = 1$ gives $3 \times 2 = 6$; $k = 2$ gives 12. So $a = 6$, $r = 2$.

why: The first term is the value at the **lower limit**, not 3.

- 2 Count the terms: k runs from 1 to 10, which is 10 terms.

why: Upper minus lower plus one.

- 3 Apply the sum formula: $S_{10} = 6 \frac{2^{10} - 1}{2 - 1} = 6 \times 1023$.

why: $2^{10} = 1024$ is worth knowing by heart.

- 4 Evaluate: 6138.

why: $6 \times 1000 = 6000$ plus $6 \times 23 = 138$.

Where it goes wrong

- Taking $a = 3$ (ignoring the $k = 1$ factor of 2), giving 3069.
- Using 2^{10} instead of $2^{10} - 1$, giving 6144.

Takeaway In $\sum ar^k$ the first term is the value at the lower limit; count terms as (upper – lower + 1).

Question 6 ALG-15 · GCSE algebra · spec M4.18

The distance-time graph of a journey is a straight line from (0, 0) to (2, 60), then horizontal to (3, 60), then a straight line to (5, 0) (time in hours, distance from home in km). What was the average speed for the whole journey?

Answer: C

Total distance $60 + 60 = 120$ km in 5 hours: 24 km/h. (Displacement is 0 but distance travelled is not.)

Full method - how and why

Average speed uses **total distance**, not net displacement - and this journey returns home.

- 1 Read the outward distance: the traveller goes from 0 to 60 km, so 60 km out.

why: The graph shows distance **from home** against time.

- 2 The horizontal section is a stop (no distance covered), and the final section returns to 0: another 60 km.

why: Coming back still counts as distance travelled.

- 3 Total distance = $60 + 60 = 120$ km over a total time of 5 hours.

why: Include the stopped hour in the time - it is part of the journey.

- 4 Average speed = $\frac{120}{5} = 24$ km/h.

why: Total distance divided by total time.

Where it goes wrong

- Using the net displacement 0 and answering 0 km/h - an offered option.
- Excluding the stopped hour, giving 30 km/h.

Takeaway Average speed = total distance $\div$ total time; stops count in the time, return legs count in the distance.

Question 7 MEN-07 · Pythagoras, mensuration and 3D · spec M5.7

A ladder of length 6.5 m leans against a vertical wall with its foot 2.5 m from the wall on horizontal ground. How far up the wall does it reach?

Answer: E

$$\sqrt{6.5^2 - 2.5^2} = \sqrt{42.25 - 6.25} = \sqrt{36} = 6.$$

Full method - how and why

A ladder against a wall is a right-angled triangle with the ladder as hypotenuse.

1 Identify the hypotenuse: the ladder, 6.5 m - the longest side, opposite the right angle at the base of the wall.
why: The wall and ground are perpendicular.

2 Apply Pythagoras: height = $\sqrt{6.5^2 - 2.5^2} = \sqrt{42.25 - 6.25}$.
why: Subtract, because we want a shorter side.

3 Compute: $\sqrt{36} = 6$ m.
why: Decimals chosen to give a whole number - a sign the working is right.

Where it goes wrong

- Adding the squares, giving $\sqrt{48.5}$ - an offered option.
- Arithmetic: $6.5^2 = 42.25$, not 36.25.

Takeaway Hypotenuse known $\Rightarrow$ subtract squares; hypotenuse unknown $\Rightarrow$ add them.

Question 8 LIN-07 · Straight lines · spec MM3.1

The line $ax + by = 12$ has x -intercept 3 and y -intercept -4 . Find $a - b$.

Answer: B

$$(3, 0): 3a = 12 \Rightarrow a = 4. (0, -4): -4b = 12 \Rightarrow b = -3. a - b = 7.$$

Full method - how and why

Intercepts are points on the line. Substitute each one to get an equation for a and for b separately.

1 x -intercept 3 means the point $(3, 0)$ lies on the line: $3a + 0 = 12$, so $a = 4$.
why: At an x -intercept, $y = 0$.

2 y -intercept -4 means $(0, -4)$ lies on it: $0 - 4b = 12$, so $b = -3$.
why: At a y -intercept, $x = 0$. The negative intercept makes b negative.

3 Compute $a - b = 4 - (-3) = 7$.
why: Subtracting a negative adds.

Where it goes wrong

- Writing $b = 3$ and answering 1.
- Mixing up which intercept is which and substituting $(0, 3)$ and $(-4, 0)$.

Takeaway An intercept is just a point with one coordinate zero - substitute it into the equation.

Question 9 BIN-06 · Binomial expansion · spec MM2.4

In the expansion of $(1+x)^n$, the coefficients of x^4 and x^5 are equal. Find n .

Answer: D

$\binom{n}{4} = \binom{n}{5}$ requires $4 + 5 = n$ (by symmetry $\binom{n}{r} = \binom{n}{n-r}$), so $n = 9$.

Full method - how and why

Two equal binomial coefficients from the same row. Symmetry gives the answer without computing factorials.

1 The condition is $\binom{n}{4} = \binom{n}{5}$.

why: In $(1+x)^n$ the coefficient of x^r is $\binom{n}{r}$.

2 Within one row of Pascal's triangle, two different entries are equal only when they are symmetric partners:

$\binom{n}{r} = \binom{n}{s}$ with $r \neq s$ forces $r + s = n$.

why: The row rises to a single peak and falls symmetrically, so equal values are mirror images.

3 Hence $n = 4 + 5 = 9$.

why: Check: $\binom{9}{4} = 126 = \binom{9}{5}$ ✓.

Where it goes wrong

- Setting up the factorial equation and mis-cancelling to get $n = 10$ or $n = 8$.
- Assuming $n = 5$ because the coefficients 'stop' there.

Takeaway Equal entries in the same row of Pascal's triangle are symmetric partners: $r + s = n$.

Question 10 BIN-01 · Binomial expansion · spec MM2.4

Find the coefficient of x^3 in the expansion of $(2+3x)^5$.

Answer: B

$$\binom{5}{3} 2^2 (3x)^3 = 10 \times 4 \times 27x^3 = 1080x^3.$$

Full method - how and why

A single coefficient from $(a+bx)^n$. Use the general term rather than expanding the whole thing.

1 Write the general term: $\binom{5}{r} 2^{5-r} (3x)^r$.

why: Every term of $(a+bx)^n$ has the form $\binom{n}{r} a^{n-r} b^r$; the powers of the two parts always add to n .

2 For x^3 take $r = 3$: $\binom{5}{3} 2^2 (3x)^3$.

why: The power of x comes only from the second bracket, so r is the power you want.

3 Evaluate each factor: $\binom{5}{3} = 10$, $2^2 = 4$, $3^3 = 27$.

why: Keep them separate - combining too early is where slips happen.

4 Multiply: $10 \times 4 \times 27 = 1080$.

why: $10 \times 108 = 1080$.

Where it goes wrong

- Forgetting the 2^2 and answering 270.
- Using $3^3 = 9$ or cubing only the x , giving 360 - both are offered.

Takeaway The general term $\binom{n}{r} a^{n-r} b^r$ carries **three** factors: the binomial coefficient and a power of each part.

Question 11 FUN-17 · Functions and transformations · spec MM8.3

The line $y = mx + c$ passes through $(2, 7)$. If m is increased by 1 and c is decreased by 2, the new line passes through $(2, k)$. Find k .

Answer: B

New value at $x = 2$: $(m + 1) \cdot 2 + (c - 2) = 2m + c = 7$. So $k = 7$.

Full method - how and why

The two changes are designed to cancel at $x = 2$. Substitute and watch the terms collapse.

- 1 From the original line: $2m + c = 7$.

why: The line passes through $(2, 7)$, so substituting gives one equation in m and c .

- 2 The new line is $y = (m + 1)x + (c - 2)$; evaluate it at $x = 2$: $2(m + 1) + (c - 2) = 2m + 2 + c - 2$.

why: Substitute $x = 2$ and expand - do not try to find m and c , which are not determined.

- 3 Simplify: $2m + c = 7$, so $k = 7$.

why: The $+2$ from the gradient change and the -2 from the intercept change cancel exactly.

Where it goes wrong

- Trying to solve for m and c individually - there is not enough information, and none is needed.
- Forgetting that the gradient change is multiplied by $x = 2$, and answering 5 or 9.

Takeaway When a question gives only one equation for two unknowns, look for a combination that is determined - here $2m + c$.

Question 12 NUM-11 · Number, units and estimation · spec M2.2

Evaluate $\left(2\frac{1}{3} - 1\frac{3}{4}\right) \div \frac{7}{6}$.

Answer: D

$$\frac{7}{3} - \frac{7}{4} = \frac{28-21}{12} = \frac{7}{12}; \frac{7}{12} \times \frac{6}{7} = \frac{1}{2}.$$

Full method - how and why

Mixed numbers and division of fractions. Convert to improper fractions, subtract, then multiply by the reciprocal.

- 1 Convert: $2\frac{1}{3} = \frac{7}{3}$ and $1\frac{3}{4} = \frac{7}{4}$.

why: Whole part times denominator plus numerator.

- 2 Subtract with a common denominator 12: $\frac{28}{12} - \frac{21}{12} = \frac{7}{12}$.

why: $\frac{7}{3} = \frac{28}{12}$ and $\frac{7}{4} = \frac{21}{12}$.

- 3 Divide by $\frac{7}{6}$ means multiply by $\frac{6}{7}$: $\frac{7}{12} \times \frac{6}{7}$.

why: Dividing by a fraction is multiplying by its reciprocal.

- 4 Cancel and multiply: the 7s cancel and $\frac{6}{12} = \frac{1}{2}$.

why: Cancel before multiplying to keep numbers small.

Where it goes wrong

- Multiplying by $\frac{7}{6}$ instead of its reciprocal, giving $\frac{49}{72}$ - an offered option.
- Subtracting the whole parts and fractions separately and mishandling the borrow.

Takeaway Improper fractions, common denominator, then multiply by the reciprocal - and cancel before multiplying.

Question 13 GRA-16 · Graphs of functions · spec MM8.7

The line $y = mx$ meets the curve $y = x^3 - x$ at three distinct points. Find the complete set of values of m .

Answer: E

$$x^3 - x - mx = x(x^2 - (1 + m)) = 0: \text{three distinct roots iff } 1 + m > 0, \text{ i.e. } m > -1.$$

Full method - how and why

Intersections of a line through the origin with a cubic. Factor out the common x and use the discriminant on what remains.

- 1 Set them equal: $mx = x^3 - x$, so $x^3 - x - mx = 0$.
why: Bring everything to one side.
- 2 Factor out x : $x(x^2 - (1 + m)) = 0$.
why: $x = 0$ is always a solution, because both graphs pass through the origin.
- 3 The other roots satisfy $x^2 = 1 + m$, which has two distinct non-zero solutions exactly when $1 + m > 0$.
why: If $1 + m = 0$ the root coincides with $x = 0$; if negative, there are no more real roots.
- 4 So $m > -1$.
why: Three distinct points in total.

Where it goes wrong

- Forgetting the root $x = 0$ and requiring three roots from the quadratic.
- Including $m = -1$, where all three roots coincide at $x = 0$.

Takeaway A line through the origin always meets $y = x^3 - x$ there; factor out x and analyse the quadratic that remains.

Question 14 DIF-02 · Differentiation · spec MM6.3

Find the coordinates of the stationary points of $y = x^3 - 3x^2 - 9x + 5$ and state their nature.

Answer: C

$$y' = 3x^2 - 6x - 9 = 3(x - 3)(x + 1). \quad y'' = 6x - 6: \text{negative at } x = -1 \text{ (max, } y = -1 - 3 + 9 + 5 = 10\text{), positive at } x = 3 \text{ (min, } y = 27 - 27 - 27 + 5 = -22\text{)}.$$

Full method - how and why

Stationary points from $f' = 0$, their nature from the sign of f'' , and the y -values by substitution.

- 1 Differentiate: $y' = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x - 3)(x + 1)$.
why: Factor out the 3 first - it makes the factorisation obvious.
- 2 Solve $y' = 0$: $x = 3$ and $x = -1$.
why: Stationary points are where the gradient vanishes.
- 3 Second derivative: $y'' = 6x - 6$. At $x = -1$, $y'' = -12 < 0$ (maximum); at $x = 3$, $y'' = 12 > 0$ (minimum).
why: Negative second derivative means the curve bends downwards - a maximum.
- 4 Find the y -values: $y(-1) = -1 - 3 + 9 + 5 = 10$; $y(3) = 27 - 27 - 27 + 5 = -22$.
why: Substitute into the original function, not the derivative.

Where it goes wrong

- Swapping the natures - for a positive cubic the maximum always comes first (smaller x).
- Substituting into y' instead of y for the coordinates.

Takeaway $f' = 0$ locates stationary points; the sign of f'' classifies them; substitute into f for the heights.

Question 15 TRI-14 · Triangles, radians and sectors · spec MM4.1

A vertical pole BT stands on horizontal ground. From a point A on the ground the angle of elevation of T is 30° ; from a point C on the ground, 20 m closer to the pole than A and in line with A and B , the angle of elevation is 60° . Find the height of the pole.

Answer: D

Let the height be h . $AB = h\sqrt{3}$ and $CB = \frac{h}{\sqrt{3}}$. $AB - CB = 20 \Rightarrow h\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right) = \frac{2h}{\sqrt{3}} = 20 \Rightarrow h = 10\sqrt{3}$.

Full method - how and why

Two right-angled triangles sharing the same height. Express both base lengths in terms of h and use the given difference.

- 1 From A : $\tan 30^\circ = \frac{h}{AB}$, so $AB = \frac{h}{\tan 30^\circ} = h\sqrt{3}$.
 why: $\tan 30^\circ = \frac{1}{\sqrt{3}}$, so dividing by it multiplies by $\sqrt{3}$.
- 2 From C : $\tan 60^\circ = \frac{h}{CB}$, so $CB = \frac{h}{\sqrt{3}}$.
 why: $\tan 60^\circ = \sqrt{3}$. The steeper angle means the shorter base - a good sanity check.
- 3 Use the 20 m gap: $AB - CB = h\sqrt{3} - \frac{h}{\sqrt{3}} = h\left(\frac{3-1}{\sqrt{3}}\right) = \frac{2h}{\sqrt{3}} = 20$.
 why: Common denominator $\sqrt{3}$: $\sqrt{3} = \frac{3}{\sqrt{3}}$.
- 4 Solve: $h = \frac{20\sqrt{3}}{2} = 10\sqrt{3}$ m.
 why: Multiply both sides by $\sqrt{3}$ and halve.

Where it goes wrong

- Adding the bases instead of subtracting (the closer point has the **shorter** base).
- Using $\tan 30^\circ = \sqrt{3}$, which swaps the two triangles.

Takeaway Two elevation angles to the same object: write each horizontal distance in terms of the height and use their difference.

Question 16 POL-16 · Polynomials, factor and remainder theorems · spec MM1.6

For which value of k does $x^2 - x + k$ divide exactly into $x^3 - 2x^2 + 3x - 4$? (Assume such a value exists.)

Answer: F

Try $x^3 - 2x^2 + 3x - 4 = (x^2 - x + k)(x + m)$: comparing x^2 : $m - 1 = -2 \Rightarrow m = -1$. Then x : $k - m = k + 1 = 3 \Rightarrow k = 2$. Constant: $km = -2 \neq -4$. No value of k works.

Full method - how and why

An 'assume it exists' question where the honest answer is that it does not. Comparing coefficients over-determines k , and the system is inconsistent.

- 1 If $x^2 - x + k$ divides exactly, the quotient is linear: $x^3 - 2x^2 + 3x - 4 = (x^2 - x + k)(x + m)$.
why: Cubic $\div$ quadratic = linear, and 'divides exactly' means no remainder term.
- 2 Compare x^2 : $m - 1 = -2$, so $m = -1$.
why: The x^2 terms are mx^2 and $-x^2$.
- 3 Compare x : $k - m = 3$, so $k = 2$.
why: The x terms are kx and $-mx$.
- 4 Check the constant term: $km = 2 \times (-1) = -2$, but the cubic has -4 . Contradiction.
why: With three equations and two unknowns the system is over-determined - the last equation is a genuine test, not a formality.
- 5 So no value of k works: the answer is 'no such value exists'.
why: The bracketed 'assume such a value exists' is a deliberate misdirection.

Where it goes wrong

- Stopping at $k = 2$ and never checking the constant term.
- Trusting the question's hint that a value exists - always verify all the coefficients.

Takeaway When comparing coefficients gives more equations than unknowns, the leftover equations must be checked; they can reveal that no solution exists.

Question 17 SEQ-19 · Sequences and series · spec MM2.3

The first three terms of a geometric sequence are x , $x + 6$ and $x + 15$. Find the fourth term.

Answer: A

Geometric: $(x + 6)^2 = x(x + 15) \Rightarrow x^2 + 12x + 36 = x^2 + 15x \Rightarrow x = 12$. The terms are 12, 18, 27 with ratio $\frac{3}{2}$, so the fourth term is $27 \times \frac{3}{2} = \frac{81}{2}$.

Full method - how and why

Three consecutive geometric terms: the middle one squared equals the product of its neighbours. That single equation finds x .

- 1 Use the geometric condition $\frac{x+6}{x} = \frac{x+15}{x+6}$, i.e. $(x+6)^2 = x(x+15)$.
why: Consecutive terms have a constant ratio, so equating the two ratios gives the condition $b^2 = ac$.
- 2 Expand: $x^2 + 12x + 36 = x^2 + 15x$.
why: The x^2 terms cancel, leaving a linear equation - which is why this method is quick.
- 3 Solve: $36 = 3x$, so $x = 12$.
why: The terms are 12, 18, 27.
- 4 Find $r = \frac{18}{12} = \frac{3}{2}$ and the fourth term $27 \times \frac{3}{2} = \frac{81}{2}$.
why: The ratio need not be an integer; leave the answer as an improper fraction.

Where it goes wrong

- Assuming the common ratio is $\frac{x+6}{x+15}$ (upside down).
- Stopping at the third term 27 or giving the ratio instead of the fourth term.

Takeaway Three terms in geometric progression satisfy $b^2 = ac$; the x^2 terms always cancel, leaving a linear equation.

Question 18 QUA-22 · Quadratic functions · spec MM1.3

The equation $x^2 + ax + b = 0$ has roots 3 and -5 . The equation $x^2 + bx + a = 0$ has roots p and q . Find $p^2 + q^2$.

Answer: D

Sum of roots $3 + (-5) = -a$, so $a = 2$; product $-15 = b$. Then $x^2 - 15x + 2 = 0$ has $p + q = 15$ and $pq = 2$, so $p^2 + q^2 = 15^2 - 2(2) = 221$.

Full method - how and why

Two linked quadratics: the roots of the first give the coefficients of the second. Sum-and-product relations do all the work.

- For $x^2 + ax + b = 0$ with roots 3 and -5 : $\text{sum} = 3 + (-5) = -2 = -a$, so $a = 2$.
why: Sum of roots $= -b/a$; with a monic quadratic that is just $-a$ - mind the double negative.
- Product $= 3 \times (-5) = -15 = b$.
why: Product of roots $= c/a = b$ here.
- The second equation is $x^2 - 15x + 2 = 0$, so $p + q = 15$ and $pq = 2$.
why: Substitute the values found, being careful that the roles of a and b swap between the two equations.
- Use $p^2 + q^2 = (p + q)^2 - 2pq = 225 - 4 = 221$.
why: The same identity as in QUA-04.

Where it goes wrong

- Getting $a = -2$ by forgetting the minus in 'sum $= -a$ ', which turns the second equation into $x^2 - 15x - 2$ and gives 229.
- Computing $(p + q)^2$ and forgetting to subtract $2pq$, giving 225.

Takeaway Read coefficients from roots with $\text{sum} = -b/a$ and $\text{product} = c/a$, and keep track of which equation you are in.

Question 19 IND-07 · Indices and surds · spec MM1.1

Which of the following is equal to $\frac{x^2 \sqrt{x}}{\sqrt[3]{x}}$ for $x > 0$?

Answer: F

The index is $2 + \frac{1}{2} - \frac{1}{3} = \frac{12+3-2}{6} = \frac{13}{6}$.

Full method - how and why

Roots disguised as indices. Convert every root to a fractional power and the question becomes adding and subtracting fractions.

- Rewrite the roots: $\sqrt{x} = x^{1/2}$ and $\sqrt[3]{x} = x^{1/3}$.
why: $\sqrt[n]{x} = x^{1/n}$. In index form the laws apply; in root form they do not.
- Combine: $\frac{x^2 \cdot x^{1/2}}{x^{1/3}} = x^{2 + \frac{1}{2} - \frac{1}{3}}$.
why: Multiplication adds indices, division subtracts them - the three operations become one sum.
- Add the fractions over a common denominator 6: $2 + \frac{1}{2} - \frac{1}{3} = \frac{12+3-2}{6} = \frac{13}{6}$.
why: 6 is the lowest common denominator of 1, 2 and 3; converting 2 to $\frac{12}{6}$ is where slips happen.

Where it goes wrong

- Adding the $\frac{1}{3}$ instead of subtracting gives $x^{17/6}$, which is offered.
- Forgetting to convert the whole number 2 into sixths.

Takeaway Turn every root into a fractional index immediately; then all you are doing is arithmetic with fractions.

Question 20 TRI-08 · Triangles, radians and sectors · spec MM4.2

The perimeter of a sector of radius r and angle θ radians is 20, and its area is 16. Find the possible values of r .

Answer: D

$2r + r\theta = 20$ and $\frac{1}{2}r^2\theta = 16$. From the first $r\theta = 20 - 2r$; then $\frac{1}{2}r(20 - 2r) = 16 \Rightarrow r^2 - 10r + 16 = 0 \Rightarrow r = 2$ or 8 .

Full method - how and why

Two sector facts, two unknowns. Eliminate θ by substitution and solve the resulting quadratic in r .

- 1** Perimeter of a sector = two radii + arc: $2r + r\theta = 20$.

why: Students often forget the two straight edges - the perimeter is not just the arc.

- 2** Rearrange for the arc: $r\theta = 20 - 2r$.

why: Keeping the product $r\theta$ together is the key, because the area formula contains it.

- 3** Area: $\frac{1}{2}r^2\theta = \frac{1}{2}r(r\theta) = \frac{1}{2}r(20 - 2r) = 16$.

why: Writing $r^2\theta$ as $r \times (r\theta)$ lets you substitute directly - no need to find θ .

- 4** Simplify and solve: $10r - r^2 = 16 \Rightarrow r^2 - 10r + 16 = 0 \Rightarrow (r - 2)(r - 8) = 0$, so $r = 2$ or 8 .

why: Both are geometrically valid: $r = 2$ gives $\theta = 8$ radians... which exceeds 2π , but the question asks only for the values of r from the equations.

Where it goes wrong

- Using perimeter = $r\theta$ alone, which gives a different quadratic.
- Finding only one root of the quadratic.

Takeaway Substitute the product $r\theta$ as a block; sector perimeter is $2r + r\theta$.